



2013 Annual Examination

FORM V

MATHEMATICS EXTENSION 1

Wednesday 28th August 2013

General Instructions

- Writing time — 3 hours
- Write using black or blue pen.
- Board-approved calculators and templates may be used.
- A list of standard integrals is provided at the end of the examination paper.

Total — 117 Marks

- All questions may be attempted.

Section I — 13 Marks

- Questions 1–13 are of equal value.
- Record your solutions to the multiple choice on the sheet provided.

Section II — 104 Marks

- Questions 14–21 are of equal value.
- All necessary working should be shown.
- Start each question in a new booklet.

Collection

- Write your name, class and master on each booklet and on your multiple choice answer sheet.
- Hand in the booklets in a single well-ordered pile.
- Hand in a booklet for each question in Section II, even if it has not been attempted.
- If you use a second booklet for a question, place it inside the first.
- Place your multiple choice answer sheet inside the answer booklet for Question Fourteen.
- Write your name and master on this question paper and submit it with your answers.

5A: DNW

5B: PKH

5C: RCF

5D: BDD

5E: KWM

5F: FMW

5G: LRP

5H: TCW

Checklist

- SGS booklets — 8 per boy
- Multiple choice answer sheet
- Candidature — 150 boys

Examiner

RCF

SECTION I - Multiple Choice

Answers for this section should be recorded on the separate answer sheet handed out with this examination paper.

QUESTION ONE

The correct factorisation of $10y^2 - 19y + 6$ is:

- (A) $(5y - 2)(2y - 3)$ (B) $(5y - 3)(2y - 2)$
 (C) $(5y - 2)(3 - 2y)$ (D) $(3 - 5y)(2 - 2y)$

QUESTION TWO

x	0	5	10
$f(x)$	1	5	9

The table of values above gives three data points from an experiment modelling an unknown function $f(x)$.

Using Simpson's rule, with three function values, to approximate $\int_0^{10} f(x) dx$, gives the answer:

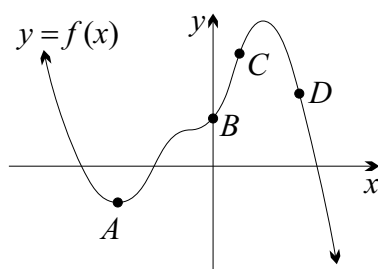
- (A) 25 (B) 50 (C) 75 (D) 100

QUESTION THREE

Given that a quadratic function with integer coefficients has a positive non-square discriminant, which of the following statements about its zeroes is true?

- (A) Equal zeroes (B) Distinct irrational zeroes
 (C) Distinct rational zeroes (D) No real zeroes

QUESTION FOUR



For which point on the graph above is $f(x) > 0$, $f'(x) > 0$ and $f''(x) < 0$?

- (A) A (B) B (C) C (D) D

Exam continues next page ...

QUESTION FIVE

A correct primitive of $2\sqrt{x}$ is:

- (A) $\frac{x\sqrt{x}}{3}$ (B) $3x\sqrt{x}$ (C) $x\sqrt{x}$ (D) $\frac{4x\sqrt{x}}{3}$

QUESTION SIX

Which of the following is not equivalent to $\log_e e^2 + \log_e \left(\frac{1}{e}\right)$?

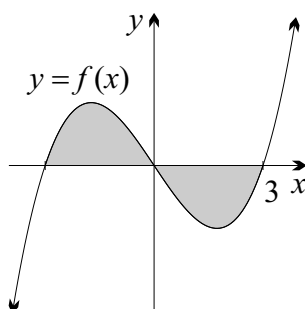
- (A) $2\log_e e - 1$ (B) $-\log_e \left(\frac{1}{e}\right)$
(C) $\log_e \left(\frac{e^3 + 1}{e}\right)$ (D) 1

QUESTION SEVEN

The derivative of $\frac{1}{(3-5x)^3}$ is:

- (A) $\frac{-15}{(3-5x)^4}$ (B) $\frac{-3}{(3-5x)^2}$ (C) $\frac{15}{(3-5x)^4}$ (D) $\frac{3}{5(3-5x)^2}$

QUESTION EIGHT



Which of the following definite integrals would correctly evaluate the area shaded above, given that $f(x)$ is an odd function?

- (A) $\int_{-3}^3 f(x) dx$ (B) $2 \int_0^3 f(x) dx$
(C) $\left| \int_{-3}^3 f(x) dx \right|$ (D) $2 \int_{-3}^0 f(x) dx$

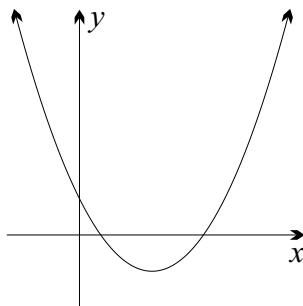
QUESTION NINE

The line perpendicular to $y = 5 - 2x$ and passing through the point $(1, -3)$ has equation:

- (A) $x - 2y - 7 = 0$ (B) $2x + y + 1 = 0$
(C) $x - 2y + 5 = 0$ (D) $2y + x + 5 = 0$

Exam continues overleaf ...

QUESTION TEN



Which of the following sets of statements is true for the quadratic $y = ax^2 + bx + c$ graphed above?

- | | |
|--------------------------------|-----------------------------------|
| (A) $a > 0, c = 0, \Delta > 0$ | (B) $a \neq 0, c > 0, \Delta < 0$ |
| (C) $a < 0, c > 0, \Delta = 0$ | (D) $a > 0, c > 0, \Delta > 0$ |

QUESTION ELEVEN

Given $a > 0$, which of the following functions is continuous but not differentiable at $x = a$?

- | | |
|-----------------------|------------------------|
| (A) $y = \log(x + a)$ | (B) $y = x - a $ |
| (C) $y = ax^3$ | (D) $y = \sqrt{x} + a$ |

QUESTION TWELVE

The quadratic equation $2x^2 - 18x + c = 0$ has one root twice the other.
What is the value of c ?

- | | | | |
|-------|-------|--------|--------|
| (A) 3 | (B) 9 | (C) 18 | (D) 36 |
|-------|-------|--------|--------|

QUESTION THIRTEEN

The derivative of $\ln\left(\frac{1}{x}\right)$ is:

- | | | | |
|--------------------|---------|----------|---------------------|
| (A) $-\frac{1}{x}$ | (B) x | (C) $-e$ | (D) $\frac{1}{x^2}$ |
|--------------------|---------|----------|---------------------|

_____ End of Section I _____

SECTION II - Written Response

Answers for this section should be recorded in the booklets provided.

Show all necessary working.

Start a new booklet for each question.

QUESTION FOURTEEN	(13 marks)	Use a separate writing booklet.	Marks
(a) Simplify:			
(i) $\log_e \left(\frac{1}{e^4} \right)$			1
(ii) $\sqrt{18} - \sqrt{8}$			1
(b) Expand and simplify:			
(i) $4 - 2(x - 3)$			1
(ii) $(2\sqrt{3} + \sqrt{5})^2$			1
(c) Find the derivative of:			
(i) $x^2 + 2x + 4$			1
(ii) $\frac{1}{x}$			1
(iii) $\log_e(2x + 1)$			1
(d) Determine the exact value of $\tan 150^\circ$.			2
(e) Find a primitive of:			
(i) $x^2 + 2x + 4$			1
(ii) $\frac{4}{x}$			1
(f) Find the limiting sum of the geometric series $3 + \frac{3}{2} + \frac{3}{4} + \dots$.			2

QUESTION FIFTEEN (13 marks) Use a separate writing booklet.

Marks

(a) Solve:

(i) $(x - 2)^2 - 3 = 0$

1

(ii) $|x - 2| = 4$

1

(iii) $(x - 2)(x + 4) > 0$

1

(b) Form the monic quadratic equation with roots 3 and -4 .

1

(c) Let $f(x) = x^3 - 8x$.

(i) Find $f(1)$, $f'(1)$ and $f''(1)$.

3

(ii) Is $f(x)$ increasing, decreasing or stationary at $x = 1$? Justify your answer.

1

(iii) Is $f(x)$ concave up or down at $x = 1$? Justify your answer.

1

(iv) Find the equation of the tangent to $y = f(x)$ at $x = 1$.

1

(d) (i) Write down the discriminant of the quadratic expression $3x^2 - mx + 3$.

1

(ii) For what values of m does the expression have no real zeroes?

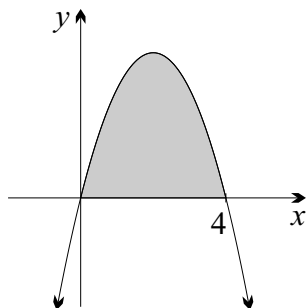
2

QUESTION SIXTEEN (13 marks) Use a separate writing booklet.**Marks**

- (a) Find the equation of the curve with derivative $\frac{dy}{dx} = 4x - 2$ that passes through the point $(2, 5)$. **2**

- (b) Evaluate $\int_1^3 2x^3 dx$ **2**

- (c) **2**



The graph above shows the parabola $y = 4x - x^2$. Calculate the area of the region enclosed between the curve and the x -axis.

- (d) Differentiate the following functions, giving your answers in a factorised form where possible:

(i) $(3x^2 + 2)^4$ **1**

(ii) $2x(x + 7)^5$ **2**

(iii) $\frac{\ln 3x}{x^2}$ **2**

- (e) Evaluate $\lim_{x \rightarrow 2} \left(\frac{x^2 - 4}{x - 2} \right)$. Show your working clearly. **2**

QUESTION SEVENTEEN (13 marks) Use a separate writing booklet.

Marks

(a) Given the sequence $\sqrt{2}, \sqrt{18}, \sqrt{50}, \dots$

(i) Show that the sequence is arithmetic.

1

(ii) Find the value of the hundredth term.

1

(iii) Find the sum of the first hundred terms.

1

(b) Suppose that $f(x) = 2x^2 - x$.

(i) Show that $f(x + h) - f(x) = 4xh + 2h^2 - h$.

1

(ii) Use the definition $f'(x) = \lim_{h \rightarrow 0} \frac{f(x + h) - f(x)}{h}$ to find $f'(x)$ from first principles.

1

(c) Consider the curve with equation $y = 3x^4 + 8x^3 + 12$.

(i) Find the coordinates of any stationary points and determine their nature.

3

(ii) Find any points of inflexion, demonstrating a change in concavity at these points.

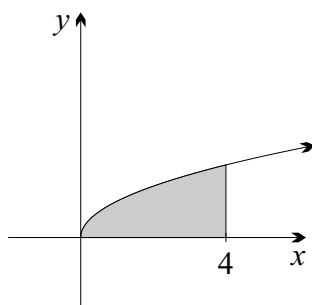
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(iii) Sketch the curve showing all the points found in parts (i) and (ii). You do NOT need to find the x -intercepts.

2

QUESTION EIGHTEEN (13 marks) Use a separate writing booklet.**Marks**

(a)

3

The diagram above shows the region enclosed by the curve $y = 3\sqrt{x}$, the x -axis and the line $x = 4$. What is the volume of the solid of revolution generated by rotating this region about the x -axis?

(b) Find the following indefinite integrals:

(i) $\int (2x + 3)^5 dx$

1

(ii) $\int \frac{x + 4}{\sqrt{x}} dx$

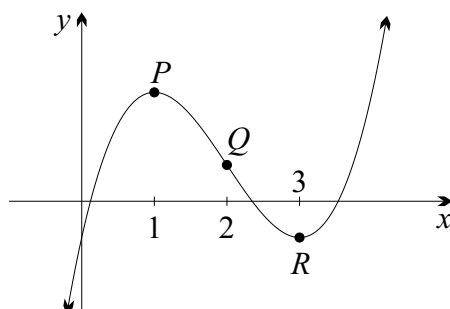
2

(iii) $\int \frac{2x}{4 + x^2} dx$

1(c) (i) Write down the equation of the line with gradient m which passes through the point $P(1, -18)$.**1**(ii) Form a quadratic equation and use the discriminant to find the values of m for which the line through P is a tangent to the parabola $y = 2x^2 + 4x - 6$.**3**(d) Find the value of k if $\int_1^k \frac{1}{x^2} dx = \frac{1}{4}$.**2**

QUESTION NINETEEN (13 marks) Use a separate writing booklet.**Marks**

(a)

2

The function $y = f(x)$ is sketched above. The points P and R are turning points and the point Q is a point of inflexion. Sketch a possible graph of the gradient function, $f'(x)$.

- (b) (i) Sketch the curve $y = \ln(x - 1)$, clearly indicating any asymptotes and any intercepts with the axes.

1

- (ii) Find the equation of the normal to $y = \ln(x - 1)$ at $x = 3$.

2

- (c) The equation $x^2 - 4x + 6 = 0$ has roots m and n .

- (i) Without solving the equation determine:

(α) $m + n$

1

(β) mn

1

(γ) $\frac{1}{m} + \frac{1}{n}$

1

- (ii) Hence, or otherwise, find a quadratic equation with integer coefficients which has roots $\frac{1}{m}$ and $\frac{1}{n}$.

2

- (d) Find the area bounded by the curve $y = x^2 - 2$, the x -axis and the lines $x = 1$ and $x = 2$.

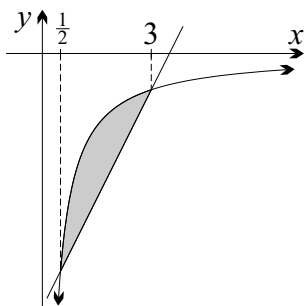
3

QUESTION TWENTY (13 marks) Use a separate writing booklet.**Marks**

- (a) Use a suitable substitution to solve
- $4^x - 5 \times 2^{x+1} + 16 = 0$
- .

3

- (b)

3

The graph above shows the line $y = 2x - 7$ and the hyperbola $y = -\frac{3}{x}$, intersecting at $x = \frac{1}{2}$ and $x = 3$.

The area of the shaded region can be written in the form $\frac{a}{4} + b \ln 6$, where a and b are integers. Find the values of a and b .

- (c) Solve
- $4 \cos(2\alpha - 45^\circ) - 2\sqrt{3} = 0$
- for the domain
- $0^\circ \leq \alpha \leq 360^\circ$
- .

3

- (d) (i) Differentiate
- $2x^3 \ln x$
- .

1

- (ii) Hence evaluate
- $\int_1^2 x^2 \ln x \, dx$
- .

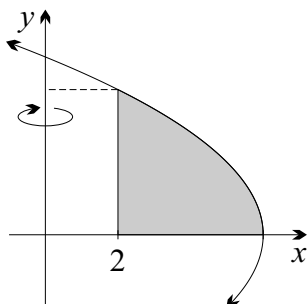
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QUESTION TWENTY ONE (13 marks) Use a separate writing booklet.

Marks

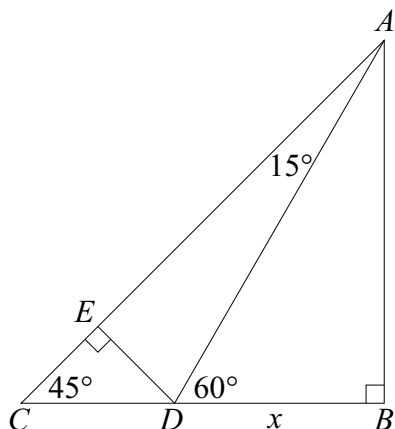
(a)

4



The diagram above shows the region bounded by the curve $4x + y^2 - 24 = 0$, the x -axis and the line $x = 2$. This region is rotated around the y -axis to create a solid of revolution. Calculate the volume of this solid.

(b)



In the diagram above, ABC is a right-angled isosceles triangle. The point D is chosen on BC so that $\angle ADB = 60^\circ$, and DE is drawn perpendicular to AC . Let $BD = x$.

(i) Show that

1

$$DE = \frac{(\sqrt{3} - 1)x}{\sqrt{2}}.$$

(ii) Hence, find an exact value for $\cos 15^\circ$.

2

- (c) (i) The function $f(t)$ satisfies $0 \leq f(t) \leq k$ for $0 \leq t \leq 1$, where k is a constant. 1

Explain using a sketch why $0 \leq \int_0^1 f(t) dt \leq k$.

- (ii) By letting $f(t) = \frac{1}{n-t} - \frac{1}{n}$, show that if $n > 1$ then 2

$$0 \leq \ln \left(\frac{n}{n-1} \right) - \frac{1}{n} \leq \frac{1}{n-1} - \frac{1}{n}.$$

- (iii) Hence show that 2

$$0 \leq \ln 2 - \sum_{n=N+1}^{2N} \frac{1}{n} \leq \frac{1}{2N}.$$

- (iv) Use the fact that $\sum_{n=6}^{10} \frac{1}{n} = \frac{1627}{2520}$ to show that 1

$$6\frac{115}{252} \leq \ln 2^{10} \leq 7\frac{115}{252}.$$

————— End of Section II —————

END OF EXAMINATION

B L A N K P A G E

B L A N K P A G E

The following list of standard integrals may be used:

$$\int x^n dx = \frac{1}{n+1} x^{n+1}, \quad n \neq -1; \quad x \neq 0, \text{ if } n < 0$$

$$\int \frac{1}{x} dx = \ln x, \quad x > 0$$

$$\int e^{ax} dx = \frac{1}{a} e^{ax}, \quad a \neq 0$$

$$\int \cos ax dx = \frac{1}{a} \sin ax, \quad a \neq 0$$

$$\int \sin ax dx = -\frac{1}{a} \cos ax, \quad a \neq 0$$

$$\int \sec^2 ax dx = \frac{1}{a} \tan ax, \quad a \neq 0$$

$$\int \sec ax \tan ax dx = \frac{1}{a} \sec ax, \quad a \neq 0$$

$$\int \frac{1}{a^2 + x^2} dx = \frac{1}{a} \tan^{-1} \frac{x}{a}, \quad a \neq 0$$

$$\int \frac{1}{\sqrt{a^2 - x^2}} dx = \sin^{-1} \frac{x}{a}, \quad a > 0, \quad -a < x < a$$

$$\int \frac{1}{\sqrt{x^2 - a^2}} dx = \ln \left(x + \sqrt{x^2 - a^2} \right), \quad x > a > 0$$

$$\int \frac{1}{\sqrt{x^2 + a^2}} dx = \ln \left(x + \sqrt{x^2 + a^2} \right)$$

NOTE : $\ln x = \log_e x, \quad x > 0$

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① $10y^2 - 19y + 6 = (5y - 2)(2y - 3)$ (A) ✓

② $\int_0^{10} f(x) dx \div \frac{1}{6} \times 10(1 + 4 \times 5 + 9)$
 $= \frac{300}{6}$
 $= 50$ (B) ✓

③ Two Distinct $\Delta > 0$ and Irrational Δ -non-square (B) ✓

④ y-co-ord positive, increasing and concave down (C) ✓

⑤ $f(x) = 2x^{\frac{1}{2}}$ $F(x) = \frac{2x^{\frac{3}{2}}}{\frac{3}{2}}$
 $= \frac{4}{3}x$ (D) ✓

⑥ $\log_e e^2 + \log_e \left(\frac{1}{e}\right) = 2 \log_e e - 1$
 $= 2 - 1 = 1$
 $-\log_e \left(\frac{1}{e}\right) = -(-1) = 1$

$\log_e \left(\frac{e^3 + 1}{e}\right) = \log_e (e^3 + 1) - \log_e e$
 $= \log_e (e^3 + 1) - 1$ (C) ✓

⑦ $y = (3 - 5x)^{-3}$
 $\frac{dy}{dx} = -3(3 - 5x)^{-4} \times (-5)$
 $= \frac{15}{(3 - 5x)^4}$ (C) ✓

⑧ $\int_{-3}^3 f(x) dx = 0$ $|\int_{-3}^3 f(x) dx| = 0$ $\int_0^3 f(x) dx < 0$

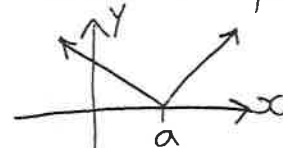
Correct Integral is $2 \int_{-3}^0 f(x) dx$ (D) ✓ $\int_3^0 f(x) dx > 0$

⑨ $y = -2x$ $m_1 = (-2)$ $m_2 = \frac{1}{2}$ $y - (-3) = \frac{1}{2}(x - 1)$
 $2y + 6 = x - 1$
 $0 = x - 2y - 7$ (A) ✓

⑩ $a > 0, \Delta > 0$ $c > 0$
 (Concave Up) (Two Distinct Zeros) y-intercept positive

(D) ✓

⑪



(B) ✓

⑫ Let roots be α and 2α
 $3\alpha = -\frac{b}{a} = \frac{18}{2}$ $2\alpha^2 = \frac{c}{a} = \frac{c}{2}$
 $\therefore \alpha = 3$ $\therefore 18 = \frac{c}{2}$
 $c = 36$ (D) ✓

⑬ $y = \ln\left(\frac{1}{x}\right)$
 $\frac{dy}{dx} = \frac{1}{\left(\frac{1}{x}\right)} \times (-x^{-2})$
 $= x \times \left(-\frac{1}{x^2}\right)$
 $= \left(-\frac{1}{x}\right)$ (A) ✓

13

⑭ a) (i) $\log_e\left(\frac{1}{e^4}\right) = (-4) \checkmark$
 (ii) $\sqrt{18} - \sqrt{8} = 3\sqrt{2} - 2\sqrt{2} = \sqrt{2} \checkmark$

c) i) $y = x^2 + 2x + 4$
 $\frac{dy}{dx} = 2x + 2 \checkmark$

(ii) $y = \frac{1}{x}$
 $= x^{-1}$
 $\frac{dy}{dx} = -x^{-2}$
 $= -\frac{1}{x^2} \checkmark$

(iii) $y = \log_e(2x+1)$
 $\frac{dy}{dx} = \frac{2}{2x+1} \checkmark$

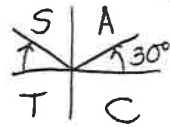
e) i) $f(x) = x^2 + 2x + 4$
 $F(x) = \frac{x^3}{3} + x^2 + 4x + C \checkmark$

(ii) $f(x) = \frac{4}{x}$
 $= 4x^{-1}$
 $F(x) = 4 \ln|x| + C \checkmark$

b) i) $4 - 2(x-3) = 4 - 2x + 6 = 10 - 2x \checkmark$

(ii) $(2\sqrt{3} + \sqrt{5})(2\sqrt{3} + \sqrt{5})$
 $= 12 + 2\sqrt{15} + 2\sqrt{15} + 5$
 $= 17 + 4\sqrt{15} \checkmark$

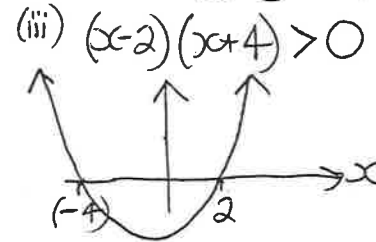
d) $\tan 150^\circ = -\tan 30^\circ \checkmark$
 $= \left(-\frac{1}{\sqrt{3}}\right) \checkmark$



f) GP $a = 3$ $r = \frac{1}{2} \checkmark$
 $S_\infty = \frac{a}{1-r} = \frac{3}{1-\frac{1}{2}} = 6 \checkmark$

⑮ a) (i) $(x-2)^2 - 3 = 0$
 $x-2 = \pm\sqrt{3}$
 $x = 2 \pm \sqrt{3} \checkmark$

(ii) $|x-2| = 4$
 $x-2 = 4$ or $x-2 = -4$
 $x = 6$ or $x = -2 \checkmark$



$x < -4$ or $x > 2 \checkmark$

b) a) $(x-\alpha)(x-\beta) = 0$
 $\therefore (x-3)(x+4) = 0$
 or $x^2 + x - 12 = 0$ } $\checkmark$ (either: must include equals zero)

c) i) $f(x) = x^3 - 8x$ $f(1) = 1 - 8 = (-7) \checkmark$
 $f'(x) = 3x^2 - 8$ $f'(1) = 3 - 8 = (-5) \checkmark$
 $f''(x) = 6x$ $f''(1) = 6 \checkmark$

(ii) $f'(1) < 0 \therefore$ Curve is decreasing at $x=1 \checkmark$

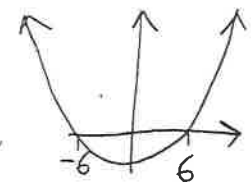
(iii) $f''(1) > 0 \therefore$ Curve is concave up at $x=1 \checkmark$

(iv) $m = (-5)$ $(1, -7)$

$\therefore$ Tangent $y + 7 = -5(x-1) \checkmark$
 $5x + y + 2 = 0$ (or $y = -5x - 2$)

d) i) $\Delta = b^2 - 4ac$
 $= (-m)^2 - 4 \times 3 \times 3$
 $= m^2 - 36 \checkmark$

(ii) No real zeroes $\Delta < 0 \checkmark$
 $\therefore (m-6)(m+6) < 0 \checkmark$
 $(-6) < m < 6 \checkmark$



(16) a) $\frac{dy}{dx} = 4x - 2$
 $y = 2x^2 - 2x + C$ ✓
 Given (2, 5)
 $5 = 2 \times 2^2 - 2 \times 2 + C$
 $5 = 8 - 4 + C$
 $1 = C$ ✓
 $\therefore y = 2x^2 - 2x + 1$

d) $\int_0^4 4x - x^3 dx = \left[2x^2 - \frac{x^4}{4} \right]_0^4$ ✓
 $= (32 - \frac{64}{4}) - 0$
 $= \frac{96}{4} - \frac{64}{4}$
 $= \frac{32}{4} = 8$ ✓
 dx) $y = (3x^2 + 2)^4$
 Chain Rule
 $\frac{dy}{dx} = 4(3x^2 + 2)^3 \times 6x$
 $= 24x(3x^2 + 2)^3$ ✓
 (ii) $y = 2x(x+7)^5$
 Product Rule
 $\frac{dy}{dx} = 2(x+7)^5 + 2x \times 5(x+7)^4$
 $= 2(x+7)^4 [x+7 + 5x]$ ✓
 $= 2(x+7)^4 (6x+7)$

e) $\lim_{x \rightarrow 2} \left(\frac{x^2 - 4}{x - 2} \right)$
 $= \lim_{x \rightarrow 2} \left(\frac{(x+2)(x-2)}{x-2} \right)$ ✓ (Working Rule)
 $= 4$ ✓

(iii) $y = \frac{\ln 3x}{x}$
 Quotient Rule
 $\frac{dy}{dx} = \frac{x^2 \times \frac{1}{3x} - \ln 3x \times 1}{x^2}$
 $= \frac{x - \ln 3x}{x^2}$
 $= \frac{1 - \ln 3x}{x^2}$ ✓

b) $\int_1^3 2x^3 dx$
 $= \left[\frac{2x^4}{4} \right]_1^3$ ✓
 $= \frac{1}{2} (3^4 - 1^4)$ ✓
 $= 40$ ✓

(17) a) $\sqrt{2}, \sqrt{18}, \sqrt{50}$
 $\sqrt{2}, 3\sqrt{2}, 5\sqrt{2}$

(i) $t_3 - t_2 = 5\sqrt{2} - 3\sqrt{2}$
 $= 2\sqrt{2}$
 $t_2 - t_1 = 3\sqrt{2} - \sqrt{2}$
 $= 2\sqrt{2}$ ✓

$\therefore$ AP $a = \sqrt{2}$ $d = 2\sqrt{2}$

(ii) $t_{100} = a + 99d$
 $= \sqrt{2} + 198\sqrt{2}$
 $= 199\sqrt{2}$ ✓

(iii) $S_{100} = \frac{100}{2} (\sqrt{2} + 199\sqrt{2})$ ✓
 $= 10000\sqrt{2}$ ✓
 $S_n = \frac{n}{2}(a + l)$

c) $g(x) = 3x^4 + 8x^3 + 12$

(i) $g'(x) = 12x^3 + 24x^2$
 $= 12x^2(x+2)$ ✓

Stat pts $g'(x) = 0$
 $\therefore x = 0$ or $x = -2$
 $g(0) = 12$ $g(-2) = 3 \times 16 - 64 + 12 = -4$

Stat pts (0, 12) and (-2, -4)

x	-3	-2	-1	0	1
g'(x)	-108	0	12	0	36

✓

$\therefore (-2, -4)$ Minimum turning point
 (0, 12) Stationary point of inflection ✓

(ii) $g''(x) = 36x^2 + 48x$
 $= 12x(3x + 4)$

b) i) $f(x) = 2x^2 - x$
 $f(x+h) = 2(x+h)^2 - (x+h)$
 $= 2(x^2 + 2xh + h^2) - x - h$
 $= 2x^2 + 4xh + 2h^2 - x - h$
 $\therefore f(x+h) - f(x) = 4xh + 2h^2 - h$ ✓

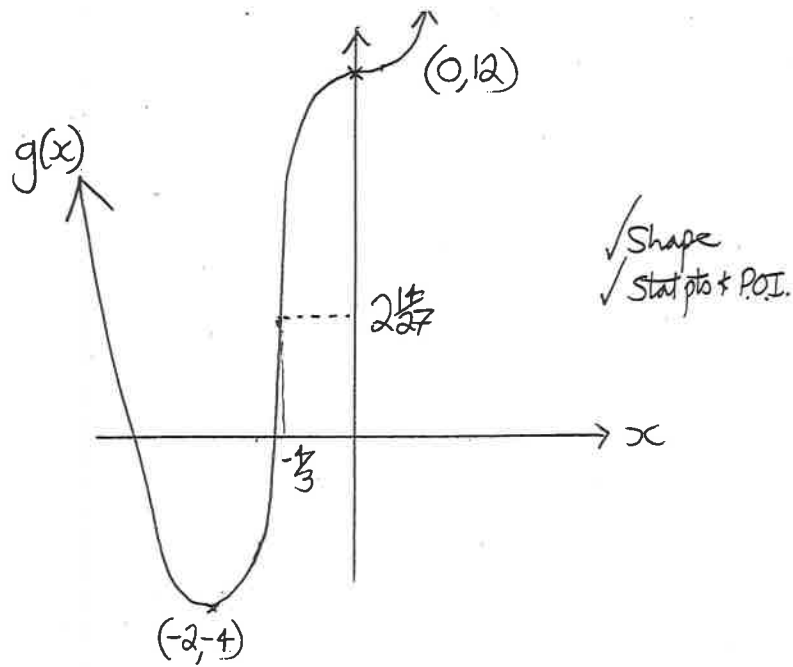
(ii) $f'(x) = \lim_{h \rightarrow 0} \left(\frac{4xh + 2h^2 - h}{h} \right)$
 $= \lim_{h \rightarrow 0} (4x + 2h - 1)$ ✓
 $= 4x - 1$

Possible pts of inflection
 $g''(x) = 0$
 $x = 0$ or $x = -\frac{4}{3}$ ✓
 $g(0) = 12$ $g(-\frac{4}{3}) = 3 \times (-\frac{4}{3})^4 + 8(-\frac{4}{3})^3 + 12$
 $= 9 \times \frac{16}{81} - 128 \times \frac{8}{27} + 12$
 $= \frac{16}{9} - \frac{1024}{27} + 12$
 $= \frac{16}{27}$ ✓

(0, 12) is stationary point of inflection from (i)

x	-2	$-\frac{4}{3}$	-1	0	1
g''(x)	48	0	-12	0	36

hence $(-\frac{4}{3}, \frac{16}{27})$ is a point of inflection too.



$$\begin{aligned}
 \textcircled{18} \text{ a) } V &= \pi \int_a^b y^2 dx \\
 &= \pi \int_0^+ q x dx \quad \checkmark \\
 &= \pi \left[\frac{q x^2}{2} \right]_0^+ \quad \checkmark \\
 &= 72\pi \text{ m}^3 \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{b) } \int_1^k \frac{1}{x^2} dx &= \int_1^k x^{-2} dx \\
 &= \left[\frac{x^{-1}}{-1} \right]_1^k \\
 &= \left[-\frac{1}{x} \right]_1^k \\
 &= -\frac{1}{k} + 1 \quad \checkmark \\
 \therefore 1 - \frac{1}{k} &= \frac{1}{4} \\
 \frac{1}{k} &= \frac{3}{4}
 \end{aligned}$$

$$\begin{aligned}
 \text{b) (i) } \int (2x+3)^5 dx &= \frac{(2x+3)^6}{6 \times 2} + C \quad k = \frac{4}{3} \quad \checkmark \\
 &= \frac{1}{12} (2x+3)^6 + C \quad \checkmark
 \end{aligned}$$

$$\begin{aligned}
 \text{(ii) } \int \frac{x+4}{\sqrt{x}} dx &= \int x^{\frac{1}{2}} + 4x^{-\frac{1}{2}} dx \quad \checkmark \\
 &= \frac{2x^{\frac{3}{2}}}{\frac{3}{2}} + 8x^{\frac{1}{2}} + C \quad \checkmark \\
 &= \frac{2}{3} (\sqrt{x})^3 + 8\sqrt{x} + C
 \end{aligned}$$

$$\text{(iii) } \int \frac{2x}{4+x^2} dx = \ln(4+x^2) + C \quad \checkmark$$

$$\begin{aligned}
 \text{c) (i) } y+18 &= m(x-1) \\
 y &= mx - m - 18 \quad \checkmark
 \end{aligned}$$

(ii) Line meets parabola $\Rightarrow$ solve simultaneously

$$2x^2 + 4x - 6 = mx - m - 18$$

$$2x^2 - mx + 4x + 12 + m = 0$$

$$2x^2 + (4-m)x + (12+m) = 0 \quad \checkmark$$

$$\Delta = (4-m)^2 - 4 \times 2 \times (12+m)$$

$$= 16 - 8m + m^2 - 96 - 8m$$

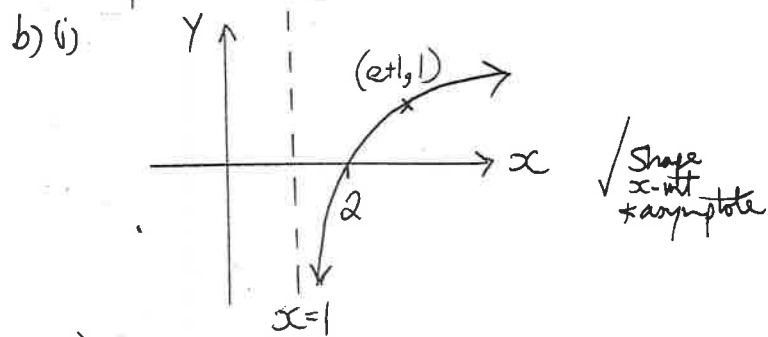
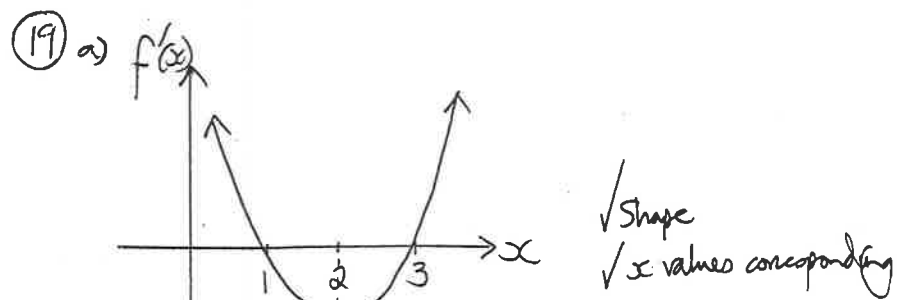
$$= m^2 - 16m - 80 \quad \checkmark$$

Tangency $\Rightarrow \Delta = 0$

$$\therefore m^2 - 16m - 80 = 0$$

$$(m-20)(m+4) = 0$$

$$m = 20 \text{ or } (-4) \quad \checkmark$$



(ii) $y = \ln(x-1)$
 $\frac{dy}{dx} = \frac{1}{x-1}$
 $\left(\frac{dy}{dx}\right)_{x=3} = \frac{1}{3-1} = \frac{1}{2}$ (3, $\ln 2$) ✓
 $m_{\text{tang}} = \frac{1}{2}$ $m_{\text{norm}} = -2$
 Eqn of normal $y - \ln 2 = -2(x-3)$ ✓
 $y + 2x - (\ln 2 + 6) = 0$ ✓
 (or equivalent)

c) $x^2 - 4x + 6 = 0$

A) $m+n = -b/a = 4$ ✓

B) $mn = c/a = 6$ ✓

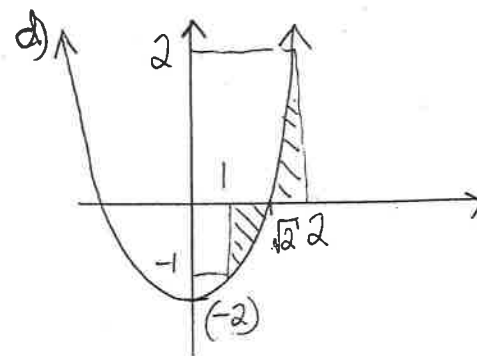
Y) $\frac{1}{m} + \frac{1}{n} = \frac{n+m}{mn} = \frac{4}{6} = \frac{2}{3}$ ✓

(ii) $\frac{1}{m} + \frac{1}{n} = \frac{2}{3}$
 $\frac{1}{m} \times \frac{1}{n} = \frac{1}{mn} = \frac{1}{6}$ ✓

$\therefore x^2 - \frac{2}{3}x + \frac{1}{6} = 0$

$6x^2 - 4x + 1 = 0$

✓ (Needs RHS = 0)

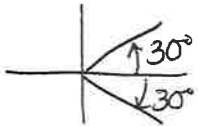


Area = $\int_1^{\sqrt{2}} x^2 - 2 dx + \int_{\sqrt{2}}^2 x^2 - 2 dx$ ✓
 $= \left[\frac{x^3}{3} - 2x \right]_1^{\sqrt{2}} + \left[\frac{x^3}{3} - 2x \right]_{\sqrt{2}}^2$
 $= \left(\frac{2\sqrt{2}}{3} - 2\sqrt{2} \right) - \left(\frac{1}{3} - 2 \right) + \left(\frac{8}{3} - 4 \right) - \left(\frac{2\sqrt{2}}{3} - 2\sqrt{2} \right)$
 $= \left| \frac{-4\sqrt{2}}{3} + \frac{5}{3} \right| + \frac{4\sqrt{2}}{3} - \frac{4}{3}$
 $= \frac{4\sqrt{2}-5}{3} + \frac{4\sqrt{2}-4}{3}$
 $= \frac{8\sqrt{2}-9}{3}$ ✓

②0 a) $4^x - 5 \times 2^{x+1} + 16 = 0$
 $(2^x)^2 - 10 \times 2^x + 16 = 0$ ✓
 $u^2 - 10u + 16 = 0$ ✓
 $(u-8)(u-2) = 0$ ✓
 $u = 8 \text{ or } 2$ ✓
 $2^x = 8 \text{ or } 2$ ✓
 $x = 3 \text{ or } 1$ ✓

c) $4 \cos(2x - 45^\circ) - 2\sqrt{3} = 0$
 $\cos(2x - 45^\circ) = \frac{\sqrt{3}}{2}$

$0^\circ \leq x \leq 360^\circ$
 $0^\circ \leq 2x \leq 720^\circ$
 $-45^\circ \leq (2x - 45^\circ) \leq 675^\circ$ ✓



$2x - 45^\circ = -30^\circ, 30^\circ, 330^\circ, 390^\circ$ ✓
 $2x = 15^\circ, 75^\circ, 375^\circ, 435^\circ$
 $x = 7\frac{1}{2}^\circ, 37\frac{1}{2}^\circ, 187\frac{1}{2}^\circ, 217\frac{1}{2}^\circ$ ✓

d) $y = 2x^3 \log x$
 $\frac{dy}{dx} = 6x^2 \log x + 2x^3 \times \left(\frac{1}{x}\right)$
 $= 2x^2(1 + 3 \log x)$
 $\frac{d}{dx}(2x^3 \log x) = 2x^2 + 6x^2 \log x$ ✓

b) Area between curves
 $A = \int_{\frac{1}{2}}^3 \left(-\frac{3}{x}\right) - (2x - 7) dx$

$= \int_{\frac{1}{2}}^3 7 - 2x - \frac{3}{x} dx$ ✓
 $= \left[7x - x^2 - 3 \ln x\right]_{\frac{1}{2}}^3$
 $= (21 - 9 - 3 \ln 3) - \left(\frac{7}{2} - \frac{1}{4} - 3 \ln \frac{1}{2}\right)$
 $= 12 - 3 \ln 3 - \frac{13}{4} + 3 \ln \frac{1}{2}$
 $= \frac{35}{4} + 3(\ln \frac{1}{2} - \ln 3)$
 $= \frac{35}{4} + 3 \ln \frac{1}{6}$
 $= \frac{35}{4} - 3 \ln 6$
 $\therefore a = 35 \quad b = (-3)$ ✓

Integrating both sides with x
 $\left[2x^3 \log x\right]_1^2 = \int_1^2 2x^2 dx + \int_1^2 6x^2 \log x dx$ ✓
 $\therefore \int_1^2 x^2 \log x dx = \frac{1}{6} \left[16 \log 2 - 2 \log 1 - \int_1^2 2x^2 dx\right]$
 $= \frac{1}{6} \left[16 \log 2 - \left[\frac{2x^3}{3}\right]_1^2\right]$ ✓
 $= \frac{1}{6} \left[16 \log 2 - \frac{16}{3} + \frac{2}{3}\right]$
 $= \frac{8}{3} \log 2 - \frac{7}{9}$ ✓

②1 a) $V = \int_0^4 \pi x^2 dy - \int_0^4 \pi 2^2 dy$ ✓ or $\int_0^4 \pi x^2 dy - \pi \times 2^2 \times 4$ (Vol of Cylindrical Hole)

$x = 6 - \frac{y^2}{4}$
 $x^2 = 36 - 3y^2 + \frac{y^4}{16}$ ✓

$V = \pi \int_0^4 \left(36 - 3y^2 + \frac{y^4}{16}\right) dy - 16\pi$
 $= \pi \left[36y - y^3 + \frac{y^5}{80}\right]_0^4 - 16\pi$ ✓

$= \left(144 - 64 + \frac{4^5}{80}\right)\pi - 16\pi$
 $= 64\pi + \frac{4^3}{5}\pi$
 $= \frac{320\pi + 64\pi}{5}$
 $= \frac{384\pi}{5} u^3$ ✓

21) at $x=2$ $y^2 = 24 - 8 = 16$ $y=4$ ($y>0$)
 a) $V = \int_0^4 \pi x^2 dy - \int_0^4 \pi 2^2 dy$ / Limits $\int_0^4 \pi x^2 dy - \pi \times 2^2 \times 4$
 (Vol of Cylindrical Hole)

$$x = 6 - \frac{y^2}{4}$$

$$x^2 = 36 - 3y^2 + \frac{y^4}{16}$$

$$V = \pi \int_0^4 \left(36 - 3y^2 + \frac{y^4}{16} \right) dy - 16\pi \quad \checkmark \text{Correct Integrand}$$

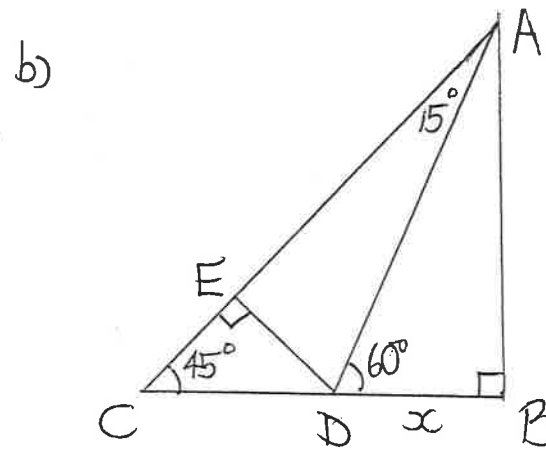
$$= \pi \left[36y - y^3 + \frac{y^5}{80} \right]_0^4 - 16\pi \quad \checkmark \text{Correct Primitive}$$

$$= \left(144 - 64 + \frac{4^5}{80} \right) \pi - 16\pi$$

$$= 64\pi + \frac{4^3}{5}\pi$$

$$= \frac{320\pi + 64\pi}{5}$$

$$= \frac{384\pi}{5} \text{ u}^3 (= 76\frac{4}{5}\pi) \quad \checkmark \text{Answer}$$



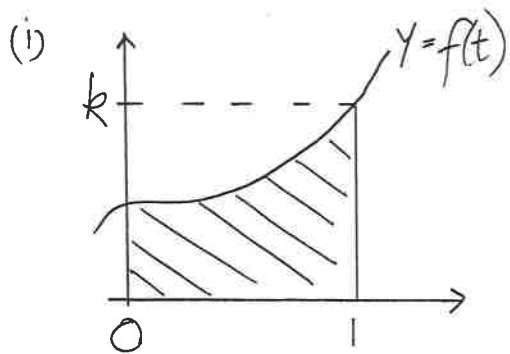
in $\triangle BAD$ $\tan 60^\circ = \frac{AB}{x}$ $\cos 60^\circ = \frac{x}{AD}$
 $\therefore AB = \sqrt{3}x$ $\therefore AD = \frac{x}{\cos 60^\circ}$
 $AD = 2x$

in $\triangle ABC$ $\angle CAB = 45^\circ$ (Angle in $\triangle CAB$)
 in $\triangle DAB$ $\angle DAB = 30^\circ$ (Angle in $\triangle DAB$)
 $\therefore \angle CAD = 15^\circ$ (Adjacent Angles)

in $\triangle ABC$ $CB = AB$ (Equal Sides of Isosceles $\triangle$)
 $\therefore CB = \sqrt{3}x$ $\therefore CD = CB - BD$ By Pythag $AC^2 = (\sqrt{3}x)^2 + (x)^2$
 $= (\sqrt{3} - 1)x$ $= 6x^2$
 $AC = \sqrt{6}x$

in $\triangle CED$ $EC = ED$ (Isosceles $\triangle$)
 By Pythag. $\therefore 2DE^2 = (\sqrt{3} - 1)^2 x^2$
 $DE^2 = \left[\frac{(\sqrt{3} - 1)}{\sqrt{2}} x \right]^2$ $\checkmark \text{Show}$
 $DE = \left(\frac{\sqrt{3} - 1}{\sqrt{2}} \right) x$

in $\triangle DEA$ $\cos 15^\circ = \frac{EA}{AD}$ $EA = AC - EC$
 $= \sqrt{6}x - \left(\frac{\sqrt{3} - 1}{\sqrt{2}} \right) x$
 $= \left(\frac{\sqrt{12} - \sqrt{3} + 1}{\sqrt{2}} \right) x = \left(\frac{\sqrt{3} + 1}{\sqrt{2}} \right) x$
 $\left(\text{or } \frac{\sqrt{2+\sqrt{3}}}{2} \right) \therefore \cos 15^\circ = \frac{\sqrt{3} + 1}{2\sqrt{2}} \quad \checkmark$



$\int_0^1 f(t) dt$ represents shaded area

Rectangular area $1 \times k = k$

$$\therefore 0 \leq \int_0^1 f(t) dt \leq k \quad \checkmark \text{ (Original + Explained)}$$

(ii) $f(t) = \frac{1}{n-t} - \frac{1}{n}$ if $n > 1$ $f(t)$ has its maximum value in the interval $0 \leq t \leq 1$ since first denominator is smallest at this point, hence $k = \frac{1}{n-1} - \frac{1}{n}$.

thus $0 \leq \int_0^1 \frac{1}{n-t} - \frac{1}{n} dt \leq \frac{1}{n-1} - \frac{1}{n}$ from (i) $\checkmark$

$$0 \leq \left[-\ln(n-t) - \frac{t}{n} \right]_0^1 \leq \frac{1}{n-1} - \frac{1}{n}$$

$$0 \leq \left(-\ln(n-1) - \frac{1}{n} \right) - \left(-\ln n - 0 \right) \leq \frac{1}{n-1} - \frac{1}{n}$$

$$0 \leq \ln n - \ln(n-1) - \frac{1}{n} \leq \frac{1}{n-1} - \frac{1}{n} \quad \checkmark$$

$$0 \leq \ln\left(\frac{n}{n-1}\right) - \frac{1}{n} \leq \frac{1}{n-1} - \frac{1}{n} \quad \blacksquare$$

(iii) Sum result (ii) from $n=N+1$ to $n=2N$

$$0 \leq \left\{ \ln\left(\frac{N+1}{N}\right) + \ln\left(\frac{N+2}{N+1}\right) + \ln\left(\frac{N+3}{N+2}\right) + \dots + \ln\left(\frac{2N}{2N-1}\right) \right. \\ \left. - \frac{1}{N+1} - \frac{1}{N+2} - \frac{1}{N+3} - \dots - \frac{1}{2N} \right\} \leq \left(\frac{1}{N} - \frac{1}{N+1} \right) + \left(\frac{1}{N+1} - \frac{1}{N+2} \right) \checkmark \\ + \dots + \left(\frac{1}{2N-1} - \frac{1}{2N} \right)$$

Telescoping the series and applying log laws.

$$0 \leq \ln\left(\frac{N+1}{N} \times \frac{N+2}{N+1} \times \dots \times \frac{2N-1}{2N-2} \times \frac{2N}{2N-1}\right) - \sum_{n=N+1}^{2N} \frac{1}{n} \leq \frac{1}{N} - \frac{1}{2N} \quad \checkmark$$

$$0 \leq \ln 2 - \sum_{n=N+1}^{2N} \frac{1}{n} \leq \frac{1}{2N} \quad \blacksquare$$

(iv) putting $N=5$.

$$0 \leq \ln 2 - \sum_{n=6}^{10} \frac{1}{n} \leq \frac{1}{10}$$

$$0 \leq \ln 2 - \frac{1627}{2520} \leq \frac{1}{10}$$

$$\frac{1627}{2520} \leq \ln 2 \leq \frac{1627}{2520} + \frac{252}{2520}$$

$$(\times 10) \quad \frac{1627}{252} \leq 10 \ln 2 \leq \frac{1879}{252} \quad \checkmark$$

$$6 \frac{115}{252} \leq \ln(2^{10}) \leq 7 \frac{115}{252} \quad \blacksquare$$